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Experimental study and thermodynamic modeling of the Al-Sc-Zr system

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ABSTRACT

As a typical system in the family of heat-resistant aluminum alloys, the Al-Sc-Zr ternary system was thermodynamically studied by experimentation and theoretical calculation. Employing the scanning electron microscopy/energy dispersive X-ray spectroscopy and X-ray diffraction, the isothermal section at 873 K was determined with equilibrated alloys. Two stable ternary compounds $Al_{75}Sc_{16}Zr_9$ and $Al_{75}Sc_{10}Zr_{15}$ were newly discovered. By means of the differential scanning calorimetry, phase transformation temperatures of selected samples were obtained. In addition, the as-cast microstructure of individual alloys was observed to identify the primary crystallization phase during solidification. Based on the experimental information mentioned above, the Al-Sc-Zr system was thermodynamically optimized combining the first-principles calculation and the CALPHAD method, with the constituent binary system Sc-Zr optimized for the first time. Using the thermodynamic parameters obtained in this work, the isothermal section, vertical section and liquidus projection were all calculated and then compared with available experimental data, most of which could be accounted for satisfactorily.

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1. Introduction

Scandium, both a rare earth element and a transition element, is known as the most outstanding alloying element in aluminum alloys. The addition of a little amount of it can significantly improve the mechanic properties at elevated temperatures [1–3], which greatly promotes the development of heat-resistant aluminum alloys and enlightens the studies afterwards on mechanisms and methods of strengthening of Al-based alloys. The strengthening effect of Sc stems from the precipitation of the Al_3Sc phase, the one possessing an $L1_2$ structure that is coherent with the matrix. These fine particles of Al_3Sc have a high thermal stability and effectively impede the motion of dislocations, grain boundaries and subgrain boundaries, thus raising the recrystallization temperature and the creep resistance. However, due to the high price of Sc, the Sc-containing aluminum alloys are mostly applied in the fields of national defense and military industry as structural materials and in the upscale civilian products. In order to expand their application, it is extremely necessary to pursue the low-cost substitutes [4–6].

Numerous studies have proved that superior mechanic properties will be obtained at high temperatures in the Sc-containing aluminum alloys where Zr is used to replace Sc partially than in the Zr-free ones [7–12]. The combined addition of Sc and Zr in the Al matrix promotes the formation of egg-shell shaped nanoparticles $L1_2$_Al_3(Sc, Zr), with the egg-part being rich in Sc and the shell-part rich in Zr [13]. The Zr-rich shell prevents the transport of the atoms due to its slow diffusion, which retards the coarsening rate and improves the thermal stability and creep resistance. In view of the importance of Sc and Zr to the heat-resistant aluminum alloys, the phase relations of the Al-Sc-Zr system need to be deeply investigated and clearly understood. With this aim, the ternary system was experimentally studied and thermodynamically optimized. A reasonable thermodynamic description was established finally. Hopefully, this work would provide guidance for upgrading current commercial heat-resistant aluminum alloys and developing novel materials.

2. Review of the Sc-Zr and Al-Sc-Zr systems

Beaudry and Daane [14] have investigated the Sc-Zr binary phase diagram by means of differential scanning calorimetry

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(DSC), with the solidus determined and the liquidus not. Only three solid solution phases exist, including liquid, bcc_A2 and hcp_A3, and not a binary compound was reported. The transformation temperature between bcc_A2 and hcp_A3 phases reaches a maximum point at ~29 at.% Zr. Based on these experimental results, Palenzona and Cirafici [15] made a review on this phase diagram, where they adjusted the melting points and transformation temperatures of pure Sc and Zr to be the values that are now widely accepted.

For the Al-Sc-Zr ternary system, the only available experimental information at present is the isothermal section at 773 K, studied by Sokolovskaya et al. [16] with equilibrated alloys. The arc-melted samples were first annealed at 773–973 K for 110 h, and then at 773 K for 400 h. The temperature for the first-order homogenizing annealing depends on the alloy compositions. According to the experimental results, there is a three-phase field consisting of Al_3Sc, Al_3Zr and Al in the Al-rich region. The Al_3Sc and Al_3Zr compounds occupy a certain area in the ternary phase diagram, for which the contents of Al vary instead of a constant value 75 at.%. With the increase of the content of Zr, the Al_2Sc phase extends into the interior of the composition triangle at a direction of reducing the Al content. Since the $AlSc_2$ and $AlZr_2$ have the same crystal structure, i.e., the Ni_2In type structure, continuous solid solution was reported to be formed. Except $AlSc_2$ and $AlZr_2$, AlZr has the largest homogeneity range, with a maximum solubility of 25.0 ± 5.0 at.% Sc, followed by Al_2Sc, where 15.5 ± 1.5 at.% Zr at most could be dissolved.

3. Experimental procedure

With 99.99 wt.% purity of Al, Sc and Zr as starting materials, twenty individual alloys, 4 g for each, were arc melted under the protection of argon. Their nominal compositions are listed in Table 1. Using Ti ball as the oxygen getter, all the alloys were melted four times and flipped after each round to ensure homogeneity. Since Sc is volatile, the composition of each sample was measured by energy dispersive X-ray spectroscopy (EDX), as is also given in Table 1. Comparing the nominal and measured alloy compositions, it can be seen that the difference becomes larger with the increase of the content of Sc, while that for the Al-rich samples is very small.

The as-melted samples were all divided into two pieces. One of them was examined by scanning electron microscopy/energy dispersive X-ray spectroscopy (SEM/EDX) to have the as-cast microstructure observed and the primary crystallization phase during solidification identified. Another piece was first sealed in the vacuumed and argon refilled quartz tube and then put in the diffusion furnace at 873 K for 1440 h, followed by a water quench. The annealed samples were cut into pieces again to be examined by SEM/EDX, X-ray diffraction (XRD) and DSC. The annealed microstructure was observed on the FEI Quanta-200 SEM and the compositions of constituent phases were analyzed by EDX, with an accelerating voltage of 20 kV and a beam spot of 5.0 or 6.0 Å. Using the Cu Kα radiation, the XRD experiments were carried out on the D/Max2500DC equipment. It scanned from 10° to 90° at a rate of 8°/min, with the voltage and the current being 40 kV and 250 mA, respectively. Bulk samples were prepared for the XRD examination when the alloys are hard and ductile, while powder ones were used when they are brittle. Ten samples, namely 1-4#, 14# and 16-20#, were selected to be tested on the NETZSCH STA 449C DSC equipment. They were heated from room temperature to 1400 °C at a rate of 10 K/min and then cooled to 200 °C at the same rate, except that the 1-3# samples were heated to 1300 °C. Both the reference and the sample were placed in the Al_2O_3 crucibles covered with a lid. During the whole heating and cooling process, argon flows through the chamber continuously to protect the samples from oxidation.

4. Calculation methods

4.1. Thermodynamic modeling

For pure elements, the Gibbs energy was adopted from the SGTE unary database [17]. The substitutional solution model was employed to describe the solution phases in the Sc-Zr and Al-Sc-Zr systems, namely liquid, hcp_A3 and bcc_A2. When establishing the thermodynamic database of the ternary system, a unified Gibbs energy expression was employed for $AlSc_2$ and $AlZr_2$ phases, since they both have the same crystal structure as Ni_2In. Other Al-Sc binary compounds, i.e., Al_3Sc, Al_2Sc and AlSc, and Al-Zr binary compounds, i.e., Al_3Zr, Al_2Zr, Al_3Zr_2, AlZr, Al_3Zr_4, Al_2Zr_3 and $AlZr_3$, have a certain solubility of the third component. Therefore, they

Table 1
Nominal compositions of individual alloys.

Alloy no.	Desired composition (at.)			EDX analyzed composition (at.)		
	Al	Sc	Zr	Al	Sc	Zr
1#	0.800	0.150	0.050	0.816	0.131	0.053
2#	0.800	0.086	0.114	0.778	0.085	0.137
3#	0.800	0.050	0.150	0.791	0.048	0.161
4#	0.691	0.270	0.039	0.726	0.217	0.057
5#	0.690	0.240	0.070	0.706	0.205	0.089
6#	0.688	0.172	0.140	0.700	0.138	0.162
7#	0.696	0.135	0.169	0.705	0.109	0.186
8#	0.631	0.102	0.267	0.638	0.084	0.278
9#	0.594	0.372	0.034	0.632	0.334	0.034
10#	0.560	0.290	0.150	0.586	0.265	0.149
11#	0.550	0.280	0.170	0.559	0.265	0.176
12#	0.445	0.203	0.352	0.507	0.143	0.350
13#	0.460	0.050	0.490	0.455	0.051	0.494
14#	0.420	0.290	0.290	0.415	0.228	0.357
15#	0.420	0.150	0.430	0.423	0.146	0.431
16#	0.376	0.362	0.262	0.387	0.246	0.367
17#	0.350	0.200	0.450	0.344	0.176	0.480
18#	0.350	0.050	0.600	0.352	0.054	0.594
19#	0.235	0.200	0.565	0.238	0.199	0.563
20#	0.180	0.050	0.770	0.182	0.055	0.763

are modeled as $Al_x(Sc,Zr)_y$, with the Gibbs energy per formula unit expressed as

$$G_{Al_x(Sc,Zr)_y} = y''_{Sc}G_{Al:Sc} + y''_{Zr}G_{Al:Zr} + yRT(y''_{Sc}\ln y''_{Sc} + y''_{Zr}\ln y''_{Zr}) + y''_{Sc}y''_{Zr}\left(\sum_{j=0,1\ldots} {}^{j}L_{Al:Sc,Zr}(y''_{Sc} - y''_{Zr})^{j}\right) \quad (1)$$

where y''_{Sc} and y''_{Zr} denote the site fraction of Sc and Zr in the second sublattice, respectively. ${}^{j}L_{Al:Sc,Zr}$ represents the j-order interaction parameter between Sc and Zr in the second sublattice. $G_{Al:Sc}$ and $G_{Al:Zr}$ could be formalized as follows.

$$G_{Al:Sc} = x^0G^{fcc}_{Al} + y^0G^{hcp}_{Sc} + A + BT \quad (2)$$

$$G_{Al:Zr} = x^0G^{fcc}_{Al} + y^0G^{hcp}_{Zr} + C + DT \quad (3)$$

For an Al-Sc binary compound, $G_{Al:Sc}$ denotes the Gibbs energy of Al_xSc_y per formula unit, and $G_{Al:Zr}$ equals to that of a fictitious phase Al_xZr_y with the structure of Al_xSc_y. The value of C, i.e., enthalpy of formation of the fictitious Al_xZr_y phase, is adopted from the first-principles calculation result and D needs to be optimized. It is a similar case for an Al-Zr binary compound.

As is evidenced by the experimental results in this work, there are two stable ternary compounds $Al_{75}Sc_{16}Zr_9$ and $Al_{75}Sc_{10}Zr_{15}$ in the Al-rich region, which were treated as stoichiometric compounds and described as follows.

$$G_{Al_xSc_yZr_z} = x^0G^{fcc}_{Al} + y^0G^{hcp}_{Sc} + z^0G^{hcp}_{Zr} + E + FT \quad (4)$$

Since the crystal structures of these two compounds are not available, the enthalpy of formation for them could not be calculated by the first-principles method. Both E and F need to be optimized in the present work.

4.2. First-principles calculation

The enthalpy of formation of the end members in the sublattice model including metastable Al-Zr and Al-Sc compounds was calculated by the VASP (Vienna Ab initio Simulation Package) [18,19] software. The interaction between ions and electrons and the exchange-correlation energy were depicted by the all-electron projector augmented wave (PAW) function [20,21] and the generalized gradient approximation (GGA) [22,23], respectively. The potential functions used in this work are Al, Sc_sv and Zr_sv. Using the Monkhorst-Pack method [24], the k-points in the Brillouin zone were produced. Both of the cutoff energy of the plane-wave and k-points were optimized until the convergence of the total energy achieved a precision of 1 meV/atom. Each structure was fully relaxed until the force between atoms was smaller than 0.01 eV/Å. Taking the metastable Al_xZr_y phase as an example, the enthalpy of formation at 0 K could be calculated by the following formula.

$$\Delta H(Al_xZr_y) = E_{total}(Al_xZr_y) - xE_{total}(Al) - yE_{total}(Zr) \quad (5)$$

where $E_{total}(Al_xZr_y)$ is the calculated total energy of Al_xZr_y with the structure of Al_xSc_y; $E_{total}(Al)$ and $E_{total}(Zr)$ are the calculated total energy of Al and Zr with the structure of fcc_A1 and hcp_A3, respectively.

In addition, the first-principles calculation was used to provide theoretic data on the thermodynamic properties when optimizing the phase diagram of the Sc-Zr binary system since the only experimental information on phase equilibrium is not enough to establish an accurate thermodynamic description. There could be several sets of thermodynamic parameters that would generate the same phase diagram optimizing with only experimental data on phase equilibria. However, the thermodynamic properties calculated by these sets of parameters could be far different. Therefore, the experimental data on phase equilibria and thermodynamic properties are both needed to help one determine a set of reliable thermodynamic parameters. Employing the special quasi-random structure (SQS) containing 16 atoms proposed by Shin et al. [25], the enthalpies of mixing for the hcp solid solutions of $Sc_{1-x}Zr_x$ (x = 0.25, 0.5, 0.75) at 0 K are expressed as

$$\Delta H(Sc_{1-x}Zr_x) = E_{total}(Sc_{1-x}Zr_x) - (1-x)E_{total}(Sc) - xE_{total}(Zr) \quad (6)$$

where $E_{total}(Sc_{1-x}Zr_x)$ is the calculated total energy of the $Sc_{1-x}Zr_x$ solid solution, and $E_{total}(Sc)$ and $E_{total}(Zr)$ are the calculated total energies of Sc and Zr with the hcp_A3 structure, respectively.

5. Results and discussion

5.1. Experimentation aspect

The equilibrium phase constitution and compositions of constituent phases determined by SEM/EDX for twenty individual alloys are summarized in Table 2. The backscattered electron images of the annealed samples 1#, 3#, 6#, 7#, 13#, 14#, 16# and 20# are presented in Fig. 1. In the Al-rich part, two ternary compounds $Al_{75}Sc_{16}Zr_9$ (τ_1) and $Al_{75}Sc_{10}Zr_{15}$ (τ_2) were found for the first time, which disapproves the existence of Al + Al_3Sc + Al_3Zr three-phase field reported by Sokolovskaya et al. [16]. Instead, three-phase fields, namely Al + Al_3Sc + τ_1, Al + Al_3Zr + τ_2 and Al + τ_1 + τ_2, and two-phase fields, namely Al + τ_1 and Al + τ_2, should exist in the Al-Al_3Sc-Al_3Zr area. One thing that should be noted is that another ternary compound Al_6ScZr (τ') was detected in the sample 1#, as is shown in Fig. 1(a). However, it was $Al_{75}Sc_{16}Zr_9$ that appeared in the samples 5# and 6#. To determine which one is stable at this temperature, the alloy with a nominal atomic composition of Al_6ScZr was additionally prepared and annealed later at 873 K for 1440 h. The actual composition of this sample measured by EDX is $Al_{77}Sc_{12}Zr_{11}$. From the SEM/EDX result, one can tell that τ_1, τ_2, Al, Al_3Zr and a little amount of Al_3Sc phases exist and no trace of τ' was found. Taking an overall consideration of the phase constitution obtained from the 1#, 5# and 6# samples and the additionally prepared one, the τ' phase is deemed as a metastable phase that can be easily formed in the Al-based alloy ($x_{Al} > 0.75$) with the atomic ratio of Sc/Zr more than 1, while the stable ternary compound should be τ_1. From the micrographs of the samples 1# and 3# shown in Fig. 1(a) and (b), three contrasts could be easily discerned, indicating a three-phase equilibrium. One can notice that the morphology of a phase has a close relation with its crystal structure, as reflected by the Al_3Sc and Al_3Zr phases in the figures. The Al_3Zr phase with a $D0_{23}$ structure is lath-shaped, which is distinct from the morphology of the Al_3Sc phase with an $L1_2$ structure. As is presented in Fig. 1(c) and (d), the samples 6# and 7# are also constituted by three equilibrium phases, with one common phase Al_2Sc that has a large solubility of Zr. The samples 13#, 14#, 16# and 20# are all in the two-phase equilibrium, as can be seen from Fig. 1(e)–(h), where the abnormally shaped phase with a black contrast is the oxide of scandium.

The XRD experiments were performed on all the annealed samples and the results corroborate the phase constitutions determined by SEM/EDX. In addition, information on crystal structures of the τ', τ_1 and τ_2 ternary compounds is revealed from the diffraction patterns. As is shown in Fig. 2, the powder diffraction files (PDFs) of Al and Al_3Sc covered all the notable peaks in sample 1#, which is constituted by three phases, namely Al, Al_3Sc and τ', according to the SEM/EDX result. This indicates that the τ' phase is characteristic of an $L1_2$ crystal structure, just like Al_3Sc. For the samples 5#, 6# and 7#, they all consist of three equilibrium phases, with either one of τ_1 and τ_2 or both of them as constituent

Table 2
Equilibrium phase constitution and compositions of constituent phases measured in this work.

Alloy no.	Phase	Al (at.)	Sc (at.)	Zr (at.)
1#	fcc_Al	0.973	0.023	0.004
	Al_3Sc	0.763	0.184	0.053
	τ'	0.750	0.132	0.118
2#	fcc_Al	0.992	0.004	0.004
	τ_2	0.744	0.101	0.155
	Al_3Zr	0.730	0.033	0.237
3#	fcc_Al	0.997	0.002	0.001
	τ_2	0.750	0.109	0.141
	Al_3Zr	0.734	0.020	0.246
4#	Al_3Sc	0.757	0.192	0.051
	Al_2Sc	0.676	0.236	0.088
5#	Al_3Sc	0.756	0.195	0.049
	τ_1	0.746	0.152	0.102
	Al_2Sc	0.659	0.207	0.134
6#	τ_1	0.750	0.162	0.088
	τ_2	0.746	0.106	0.148
	Al_2Sc	0.659	0.161	0.180
7#	τ_2	0.744	0.098	0.158
	Al_3Zr	0.736	0.045	0.219
	Al_2Sc	0.656	0.124	0.220
8#	Al_2Sc	0.649	0.093	0.258
	Al_2Zr	0.640	0.043	0.317
9#	Al_2Sc	0.676	0.264	0.060
	AlSc	0.523	0.464	0.013
10#	Al_3Zr_2	0.588	0.253	0.159
	AlSc	0.493	0.433	0.074
11#	Al_3Zr_2	0.581	0.244	0.175
	AlSc	0.500	0.408	0.092
12#	Al_3Zr_2	0.569	0.132	0.299
	AlZr	0.499	0.126	0.375
13#	AlZr	0.486	0.041	0.473
	Al_3Zr_4	0.432	0.031	0.537
14#	Al_3Zr_4	0.435	0.164	0.401
	Al_2Zr_3	0.410	0.270	0.320
15#	Al_3Zr_4	0.435	0.095	0.470
	Al_2Zr_3	0.405	0.272	0.323
16#	Al_2Zr_3	0.403	0.177	0.420
	$AlZr_2$	0.366	0.306	0.328
17#	Al_2Zr_3	0.397	0.109	0.494
	$AlZr_2$	0.338	0.184	0.478
18#	Al_2Zr_3	0.394	0.035	0.571
	$AlZr_2$	0.332	0.061	0.607
19#	$AlZr_2$	0.302	0.167	0.531
	$AlZr_3$	0.239	0.143	0.618
	hcp_Zr	0.161	0.259	0.580
20#	$AlZr_3$	0.222	0.048	0.730
	hcp_Zr	0.114	0.086	0.800

phases, based on the results of SEM/EDX. Since there are no PDFs for these two ternary compounds, only three phases Al_3Sc, Al_3Zr and Al_2Sc could be determined. Comparing the XRD patterns of the samples 5#, 6# and 7# in Fig. 3, it can be noticed that results of the first two are almost the same, with a little difference from that of the last one. For the samples 5# and 6#, except the diffraction peaks that could be fitted by the PDF of Al_3Sc, there are two little peaks at 31.4° and 53.0° showing up at the same position as those from the PDF of Al_3Zr, while other six distinct peaks at 22.7, 27.0, 45.1, 63.1, 65.7 and 78.4° in the Al_3Zr PDF do not appear. Therefore, it can be inferred that τ_1 and τ_2 have a crystal structure much closer to that of Al_3Sc than Al_3Zr.

Compared with the isothermal section at 773 K measured by Sokolovskaya et al. [16], the isothermal section at 873 K in this work is far different. Usually, two isothermal sections with a difference of 100 K in temperature would differentiate in partial phase relations and homogeneity ranges of some binary compounds. This large difference in phase relations could not be explained sufficiently by the temperature difference and is supposed to mainly originate from the heating history of individual alloys. In this work, the samples were all annealed at 873 K for 1440 h, much longer than 510 h in total for the two-order annealing carried out in the work of Sokolovskaya et al. [16]. From this point of view, the experimental data obtained in this work should be more reliable.

The DSC curves of heating and cooling for the samples 3#, 14# and 16# are presented in Fig. 4, where the red lines represent heating and the blue ones cooling. The phase transformation temperatures are determined from the onset points of the endothermic peaks during heating and the end points of the exothermic peaks during cooling, except that the peak temperature are used for the last thermal event in the heating curve of the sample 14#, as shown in Fig. 4(b), and for the liquidus temperature of the sample 19#. It was found that the transformation temperatures determined from the heating and cooling curves agree with each other very well for the samples in the Al-rich corner, while it is not the case for those containing more Sc and Zr. This can be seen from Fig. 4. Since there are Sc and Zr elements in the samples, the oxidation could not be avoided when heating to high temperatures. Therefore, the phase transformation temperatures determined from the heating curves are more reliable, which are listed in Table 3.

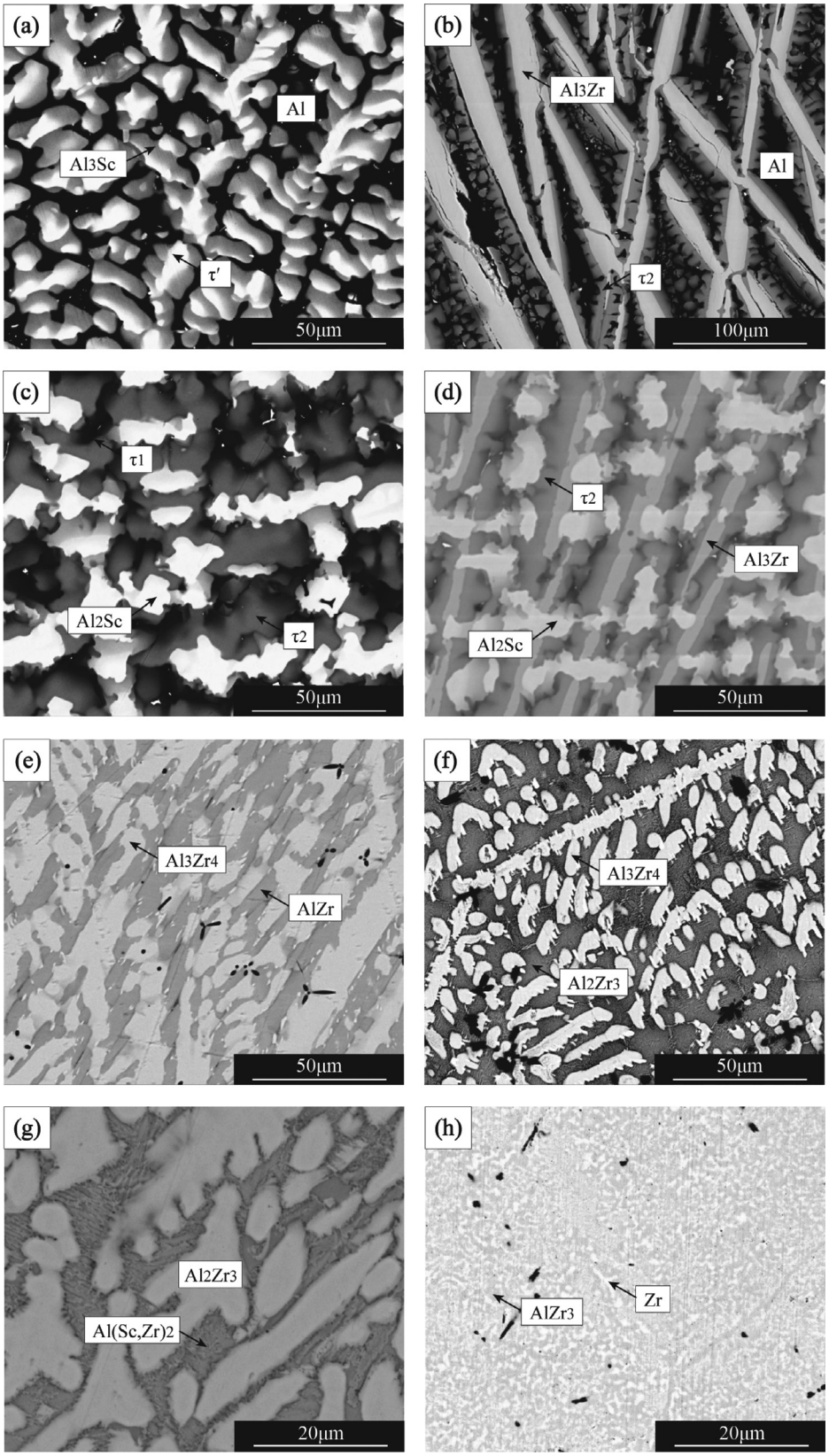

Fig. 1. Backscattered electron images of the annealed samples: (a) 1#; (b) 3#; (c) 6#; (d) 7#; (e) 13#; (f) 14#; (g) 16#; (h) 20#.

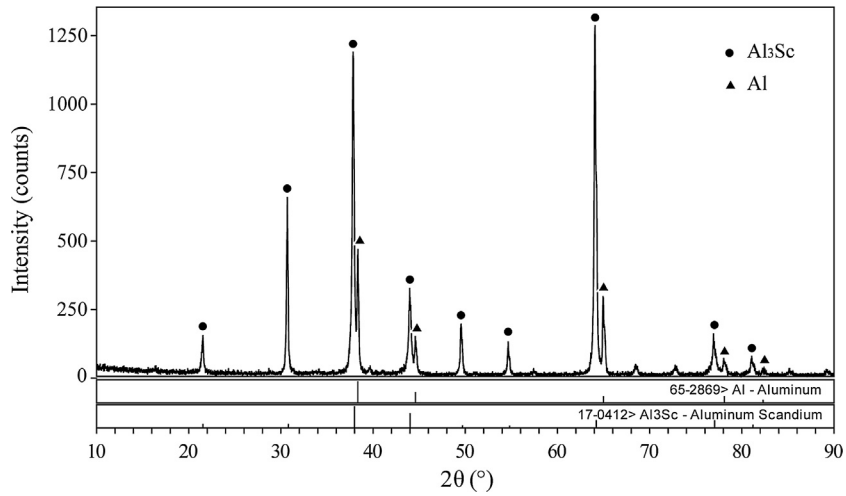

Fig. 2. XRD pattern of the annealed sample 1#.

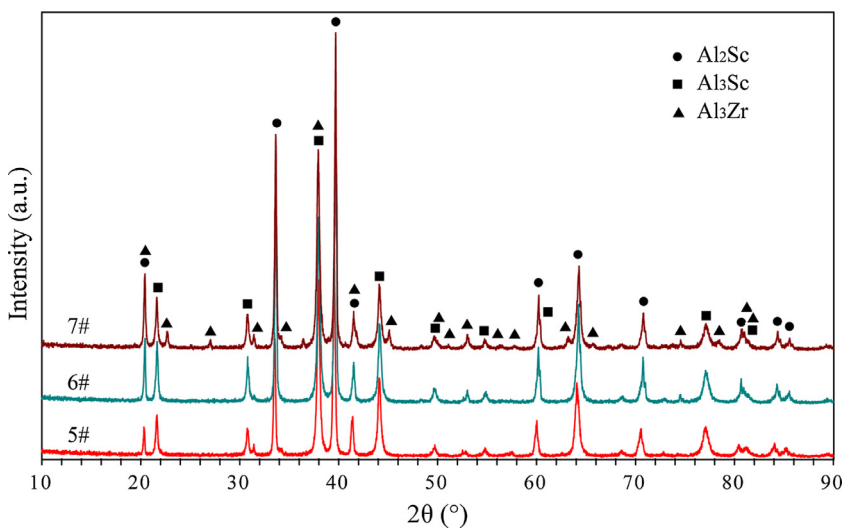

Fig. 3. XRD patterns of the annealed samples 5#, 6# and 7#.

5.2. Modeling aspect

Based on the phase diagram reviewed by Palenzona and Cirafici [15] and the enthalpy of mixing of the $Sc_{1-x}Zr_x$ (x = 0.25, 0.5, 0.75) hcp_A3 solid solution calculated in this work, the Sc-Zr binary system was optimized on the Parrot module of the Thermo_calc software developed by Sundman et al. [26]. The optimization started with the hcp_A3 phase, which has more related information including both experimental and theoretical data, followed by the bcc_A2 phase and the liquid phase.

The enthalpy of mixing of the hcp_A3 solid solution was calculated using the optimized thermodynamic parameters, as is shown in Fig. 5. Good agreement could be found between the thermodynamic calculation result and the first-principles calculation data. Fig. 6 presents the calculated phase diagram of the Sc-Zr system. The data designated by the triangle were digitized from the phase diagram reviewed by Palenzona and Cirafici [15]. It can be seen that the calculated phase diagram agrees very well with the reviewed one.

For Al-Sc-Zr ternary system, the optimization was based on the experimental information in this work as well as the enthalpy of formation of metastable binary compounds calculated by the first-principles method. The thermodynamic parameters of the Al-Sc and Al-Zr binary systems were adopted from our previous work [27] and the work of Wang et al. [28], respectively. Table 4 summarizes the optimized thermodynamic parameters of this ternary system.

The calculated isothermal section of the Al-Sc-Zr system at 873 K is presented in Fig. 7, which accounts for the measured phase relations satisfactorily. It should be noted that the maximum solubility of Sc in the Al_3Zr_4 and Al_2Zr_3 phases were not determined in this work. The sample 14# was supposed to be in the AlZr + Al_3Zr_4 + $AlSc_2$ three-phase field. However, it fell into the Al_3Zr_4 + Al_2Zr_3 two-phase field because of the volatilization of Sc. Therefore, the homogeneity ranges of these two compounds were optimized following the rule that all the experimental information including data on equilibrium phase relations, phase transformation temperatures and primary crystallization phases can be explained as well as possible. With the optimized thermodynamic parameters, the phase fields that are lack of experimental data could be predicted. It should be noted that a large difference exists between the calculated and measured composition of the constituent hcp_A3 phase for the samples 19# and 20#, since the size of this phase is too small (<1μ) for the EDX to produce accurate results. Therefore,

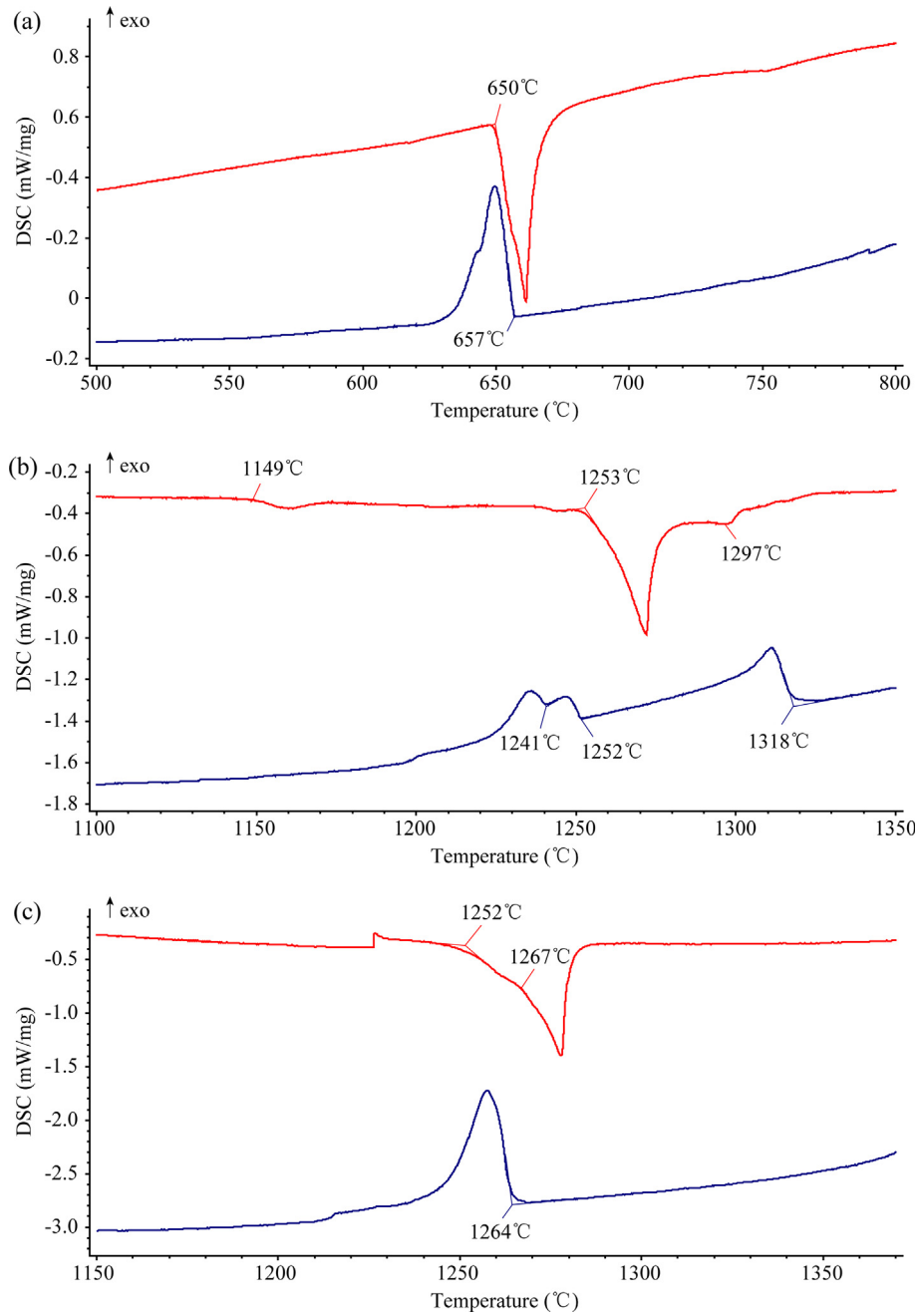

Fig. 4. DSC heating and cooling curves of the annealed samples: (a) 3#; (b) 14#; (c) 16#.

the tie-line of the sample 20# and the two sides of the tie-triangle of the sample 19# measured in this work are plotted with dashed lines.

The phase transformation temperatures of the samples by calculation are compared with those measured by DSC in Table 3, where the calculated invariant reactions and phase boundaries corresponding to the transition temperatures are also listed. When interpreting the DSC signals, vertical sections that contain the composition of each sample were calculated respectively and the experimental data from DSC were superimposed for comparison. The assignment of the invariant reaction or phase boundary to each measured temperature is tentative. Only the calculated invariant reaction for the samples 2# and 3# can be confirmed by the microstructure of the annealed samples since it takes place at about 650 °C which is close to the annealing temperature. Although a generally good agreement is achieved between the calculated and measured temperatures, there are still some calculated reactions with the eutectoid or peritectoid type that were not detected by the DSC experiments. This is understandable because the reactions with only solid phases taking part in are very difficult to take place and the corresponding thermal effects could be very weak. Fig. 8 presents the calculated vertical section at 5 at.% Sc, along with the DSC data of the samples 18# and 3#. The measured phase transformation temperatures during heating and cooling are designated by upward and downward triangles, respectively. According to the calculation result, the sample 3# only experienced one invariant reaction Liquid + Al_3Zr ↔ τ_2 + fcc_A1 at 661 °C below the temperature of 1300 °C, while the measured temperatures by DSC during heating and cooling are 650 °C and 657 °C, respectively, which agree well with the calculation result. As for

Table 3
Measured and calculated phase transformation temperatures and corresponding calculated invariant reactions or phase boundaries.

Alloy no.	Measured temperature (°C)	Calculated temperature (°C)	Invariant reaction or phase boundary
1#	–	1128	Liquid + τ_1/Liquid + τ_1 + Al_3Sc
	650	660	Liquid + $\tau_1 \leftrightarrow$ fcc_A1 + Al_3Sc
2#	650	661	Liquid + $Al_3Zr \leftrightarrow \tau_2$ + fcc_A1
3#	650	661	Liquid + $Al_3Zr \leftrightarrow \tau_2$ + fcc_A1
4#	1327	1358	Liquid + τ_1 + $Al_2Sc \leftrightarrow Al_3Sc$
	907	899	τ_1 + Al_2Sc + Al_3Sc/Al_2Sc + Al_3Sc
14#	1297	1333	AlSc + Al_2Zr_3/AlSc + Al_2Zr_3 + Al_3Zr_4
	1253	–	–
	1149	1118	AlSc + Al_2Zr_3 + Al_3Zr_4/Al_2Zr_3 + Al_3Zr_4
16#	1267	1321	Liquid + Al_2Zr_3/Liquid + Al_2Zr_3 + $Al(Sc,Zr)_2$
	1252	1309	Liquid + Al_2Zr_3 + $Al(Sc,Zr)_2$/Al_2Zr_3 + $Al(Sc,Zr)_2$
17#	1300	1273	Liquid + Al_2Zr_3/Liquid + Al_2Zr_3 + bcc_A2
	1267	1268	Liquid + Al_2Zr_3 + bcc_A2/Al_2Zr_3 + bcc_A2
	–	1075	Al_2Zr_3 + bcc_A2/Al_2Zr_3 + bcc_A2 + hcp_A3
	–	1074	Al_2Zr_3 + bcc_A2 + hcp_A3/Al_2Zr_3 + hcp_A3
	–	972	Al_2Zr_3 + hcp_A3/Al_2Zr_3 + hcp_A3 + $Al(Sc,Zr)_2$
	909	934	Al_2Zr_3 + hcp_A3 + $Al(Sc,Zr)_2$/Al_2Zr_3 + $Al(Sc,Zr)_2$
18#	1329	1334	Liquid + Al_2Zr_3/Liquid + Al_2Zr_3 + bcc_A2
	1314	1333	Liquid + Al_2Zr_3 + bcc_A2/Al_2Zr_3 + bcc_A2
	1263	–	–
	–	1045	Al_2Zr_3 + bcc_A2/Al_2Zr_3 + bcc_A2 + $AlZr_3$
	–	1031	Al_2Zr_3 + bcc_A2 + $AlZr_3$/Al_2Zr_3 + $AlZr_3$
	–	958	Al_2Zr_3 + $AlZr_3$/Al_2Zr_3 + $AlZr_3$ + $Al(Sc,Zr)_2$
	909	947	Al_2Zr_3 + $AlZr_3$ + $Al(Sc,Zr)_2$/Al_2Zr_3 + $Al(Sc,Zr)_2$
19#	1328	1330	Liquid/Liquid + bcc_A2
	–	1285	Liquid + bcc_A2/Liquid + bcc_A2 + Al_2Zr_3
	–	1280	Liquid + bcc_A2 + Al_2Zr_3/bcc_A2 + Al_2Zr_3
	–	1064	bcc_A2 + Al_2Zr_3/bcc_A2 + Al_2Zr_3 + hcp_A3
	1019	1061	bcc_A2 + Al_2Zr_3 + hcp_A3/Al_2Zr_3 + hcp_A3
	–	919	Al_2Zr_3 + hcp_A3/Al_2Zr_3 + hcp_A3 + $Al(Sc,Zr)_2$
	914	916	Al_2Zr_3 + hcp_A3 + $Al(Sc,Zr)_2$/hcp_A3 + $Al(Sc,Zr)_2$
	–	629	hcp_A3 + $Al(Sc,Zr)_2$/hcp_A3 + $Al(Sc,Zr)_2$ + $AlZr_3$
20#	1130	1169	bcc_A2/bcc_A2 + Al_2Zr_3
	–	1032	bcc_A2 + Al_2Zr_3/bcc_A2 + Al_2Zr_3 + $AlZr_3$
	1006	1017	bcc_A2 + Al_2Zr_3 + $AlZr_3$/bcc_A2 + $AlZr_3$
	–	981	bcc_A2 + $AlZr_3$/bcc_A2 + $AlZr_3$ + hcp_A3
	949	975	bcc_A2 + $AlZr_3$ + hcp_A3/$AlZr_3$ + hcp_A3

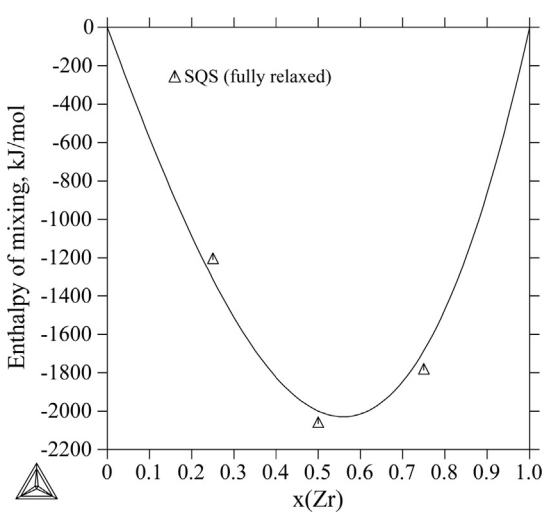

Fig. 5. Calculated enthalpy of mixing of the hcp_A3 solid solution of the Sc-Zr system along with the first-principles calculation results.

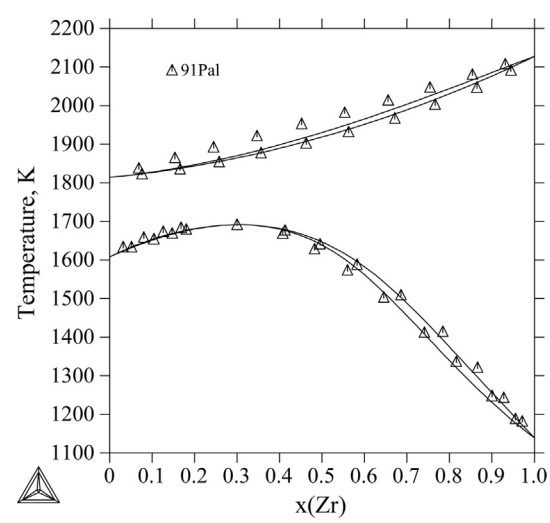

Fig. 6. Calculated phase diagram of the Sc-Zr binary system.

the sample 18#, the agreement between the calculated and measured temperatures is not as well as that for the sample 3#. For the reaction Al_2Zr_3 + hcp_A3 $\leftrightarrow$ $Al(Sc,Zr)_2$, the deviation is as large as 38 °C. The peak on the DSC curve corresponding to this reaction is a little broad and not in good shape, which may has an influence on the analyzed result.

With the thermodynamic parameters optimized in this work, the liquidus projection of the Al-Sc-Zr system was calculated, as is presented in Fig. 9. Five regions in the Al-rich part are enlarged so as to demonstrate the invariant reactions clearly. Among the twenty individual alloys that were produced, eight different primary crystallization phases were found, namely τ' for 1#, τ_2 for

Table 4
Thermodynamic parameters of the Al-Sc-Zr system.

Phase	Thermodynamic parameters	Ref.
Liquid: (Al, Sc, Zr)	${}^{0}L^{Liq}_{Al,Sc} = -131,778 + 3.639T$	[27]
	${}^{1}L^{Liq}_{Al,Sc} = -20,126 + 8.877T$	[27]
	${}^{2}L^{Liq}_{Al,Sc} = 50,985 - 24.917T$	[27]
	${}^{0}L^{Liq}_{Al,Zr} = -207,140 + 42.911T$	[28]
	${}^{1}L^{Liq}_{Al,Zr} = -22,643 + 11.442T$	[28]
	${}^{2}L^{Liq}_{Al,Zr} = 56,610 - 7.807T$	[28]
	${}^{0}L^{Liq}_{Sc,Zr} = 18,800 - 11T$	This work
	${}^{2}L^{Liq}_{Sc,Zr} = -20,124 + 10.8T$	This work
fcc_A1: $(Al, Sc, Zr)_1(Va)_1$	${}^{0}L^{fcc_A1}_{Al,Sc:Va} = -109,500 + 2.5T$	[27]
	${}^{0}L^{fcc_A1}_{Al,Zr:Va} = -152,947 + 21.3T$	[28]
bcc_A2: $(Al, Sc, Zr)_1(Va)_3$	${}^{0}L^{bcc_A2}_{Al,Sc:Va} = -113,589$	[27]
	${}^{0}L^{bcc_A2}_{Al,Zr:Va} = -165,348 + 13.235T$	[28]
	${}^{1}L^{bcc_A2}_{Al,Zr:Va} = -36,570$	[28]
	${}^{2}L^{bcc_A2}_{Al,Zr:Va} = 16,806$	[28]
	${}^{0}L^{bcc_A2}_{Sc,Zr:Va} = 10,000 - 5.22T$	This work
hcp_A3: $(Al, Sc, Zr)_1(Va)_{0.5}$	${}^{0}L^{hcp_A3}_{Al,Sc:Va} = -106,397$	[27]
	${}^{0}L^{hcp_A3}_{Al,Zr:Va} = -166,254 + 13.235T$	[28]
	${}^{1}L^{hcp_A3}_{Al,Zr:Va} = -36,570$	[28]
	${}^{2}L^{hcp_A3}_{Al,Zr:Va} = 16,806$	[28]
	${}^{0}L^{hcp_A3}_{Sc,Zr:Va} = 8500 - 5.77T$	This work
Al_3Sc: $(Al)_3(Sc, Zr)_1$	$G^{Al_3Sc}_{Al:Sc} = -156,700 + 7.2T + 3{}^{0}G^{fcc}_{Al} + {}^{0}G^{hcp}_{Sc}$	[27]
	$G^{Al_3Sc}_{Al:Zr} = -177,627 + 23.25T + 3{}^{0}G^{fcc}_{Al} + {}^{0}G^{hcp}_{Zr}$	This work
	${}^{0}L^{Al_3Sc}_{Al:Sc,Zr} = -84,160$	This work
Al_2Sc: $(Al)_2(Sc, Zr)_1$	$G^{Al_2Sc}_{Al:Sc} = -142,500 + 8.8T + 2{}^{0}G^{fcc}_{Al} + {}^{0}G^{hcp}_{Sc}$	[27]
	$G^{Al_2Sc}_{Al:Zr} = -153,992 + 24.85T + 2{}^{0}G^{fcc}_{Al} + {}^{0}G^{hcp}_{Zr}$	This work
	${}^{0}L^{Al_2Sc}_{Al:Sc,Zr} = -150,000 + 5T$	This work
	${}^{1}L^{Al_2Sc}_{Al:Sc,Zr} = 70,000$	This work
AlSc: $(Al)_1(Sc, Zr)_1$	$G^{AlSc}_{Al:Sc} = -94,100 + 6.4T + {}^{0}G^{fcc}_{Al} + {}^{0}G^{hcp}_{Sc}$	[27]
	$G^{AlSc}_{Al:Zr} = -55,627 - 2.72T + {}^{0}G^{fcc}_{Al} + {}^{0}G^{hcp}_{Zr}$	This work
	${}^{0}L^{AlSc}_{Al:Sc,Zr} = -100,000$	This work
$Al(Sc, Zr)_2$: $(Al)_1(Sc, Zr)_2$	$G^{Al(Sc,Zr)_2}_{Al:Sc} = -112,484 + 1.6T + {}^{0}G^{fcc}_{Al} + 2{}^{0}G^{hcp}_{Sc}$	[27]
	$G^{Al(Sc,Zr)_2}_{Al:Zr} = -145,074 + 20.476T + {}^{0}G^{fcc}_{Al} + 2{}^{0}G^{hcp}_{Zr}$	[28]
	${}^{0}L^{Al(Sc,Zr)_2}_{Al:Sc,Zr} = -18,000$	This work
$AlZr_3$: $(Al)_1(Zr, Sc)_3$	$G^{AlZr_3}_{Al:Zr} = -144,652 + 17.684T + {}^{0}G^{fcc}_{Al} + 3{}^{0}G^{hcp}_{Zr}$	[28]
	$G^{AlZr_3}_{Al:Sc} = -98,820 + 14T + {}^{0}G^{fcc}_{Al} + 3{}^{0}G^{hcp}_{Sc}$	This work
	${}^{0}L^{AlZr_3}_{Al:Zr,Sc} = -40,600$	This work
Al_3Zr_5: $(Al)_3(Zr)_5$	$G^{Al_3Zr_5}_{Al:Zr} = -411,872 + 45.992T + 3{}^{0}G^{fcc}_{Al} + 5{}^{0}G^{hcp}_{Zr}$	[28]
Al_2Zr_3: $(Al)_2(Zr, Sc)_3$	$G^{Al_2Zr_3}_{Al:Zr} = -275,900 + 33.67T + 2{}^{0}G^{fcc}_{Al} + 3{}^{0}G^{hcp}_{Zr}$	[28]
	$G^{Al_2Zr_3}_{Al:Sc} = -173,189 + 25T + 2{}^{0}G^{fcc}_{Al} + 3{}^{0}G^{hcp}_{Sc}$	This work
	${}^{0}L^{Al_2Zr_3}_{Al:Zr,Sc} = -150,500$	This work
Al_3Zr_4: $(Al)_3(Zr, Sc)_4$	$G^{Al_3Zr_4}_{Al:Zr} = -409,360 + 57.652T + 3{}^{0}G^{fcc}_{Al} + 4{}^{0}G^{hcp}_{Zr}$	[28]
	$G^{Al_3Zr_4}_{Al:Sc} = -218,246 + 28T + 3{}^{0}G^{fcc}_{Al} + 4{}^{0}G^{hcp}_{Sc}$	This work
	${}^{0}L^{Al_3Zr_4}_{Al:Zr,Sc} = -259,000$	This work
Al_4Zr_5: $(Al)_4(Zr)_5$	$G^{Al_4Zr_5}_{Al:Zr} = -498,816 + 47.88T + 4{}^{0}G^{fcc}_{Al} + 5{}^{0}G^{hcp}_{Zr}$	[28]
AlZr: $(Al)_1(Zr, Sc)_1$	$G^{AlZr}_{Al:Zr} = -129,900 + 22.028T + {}^{0}G^{fcc}_{Al} + {}^{0}G^{hcp}_{Zr}$	[28]
	$G^{AlZr}_{Al:Sc} = -80,166 + 5.76T + {}^{0}G^{fcc}_{Al} + {}^{0}G^{hcp}_{Sc}$	This work
	${}^{0}L^{AlZr}_{Al:Zr,Sc} = -56,000$	This work
Al_3Zr_2: $(Al)_3(Zr, Sc)_2$	$G^{Al_3Zr_2}_{Al:Zr} = -276,615 - 139.15T + 21.645T\ln T + 3{}^{0}G^{fcc}_{Al} + 2{}^{0}G^{hcp}_{Zr}$	[28]
	$G^{Al_3Zr_2}_{Al:Sc} = -208,940 + 25T + 3{}^{0}G^{fcc}_{Al} + 2{}^{0}G^{hcp}_{Sc}$	This work
	${}^{0}L^{Al_3Zr_2}_{Al:Zr,Sc} = -229,500$	This work
Al_2Zr: $(Al)_2(Zr, Sc)$	$G^{Al_2Zr}_{Al:Zr} = -153,798 - 89.178T + 13.251T\ln T + 2{}^{0}G^{fcc}_{Al} + {}^{0}G^{hcp}_{Zr}$	[28]
	$G^{Al_2Zr}_{Al:Sc} = -139,677 + 10.56T + 2{}^{0}G^{fcc}_{Al} + {}^{0}G^{hcp}_{Sc}$	This work
	${}^{0}L^{Al_2Zr}_{Al:Zr,Sc} = -93,960$	This work
Al_3Zr: $(Al)_3(Zr, Sc)_1$	$G^{Al_3Zr}_{Al:Zr} = -189,524 - 97.492T + 15.576T\ln T + 3{}^{0}G^{fcc}_{Al} + {}^{0}G^{hcp}_{Zr}$	[28]
	$G^{Al_3Zr}_{Al:Sc} = -160,120 + 22.34T + 3{}^{0}G^{fcc}_{Al} + {}^{0}G^{hcp}_{Sc}$	This work
	${}^{0}L^{Al_3Zr}_{Al:Zr,Sc} = -53,040$	This work

Table 4 (*continued*)

Phase	Thermodynamic parameters	Ref.
τ_1: $(Al)_{75}(Sc)_{16}(Zr)_9$	$G^{\tau_1}_{Al:Sc:Zr} = -4,509,700 + 20T + 75^0G^{fcc}_{Al} + 16^0G^{hcp}_{Sc} + 9^0G^{hcp}_{Zr}$	This work
τ_2: $(Al)_{75}(Sc)_{10}(Zr)_{15}$	$G^{\tau_2}_{Al:Sc:Zr} = -4,900,000 + 190T + 75^0G^{fcc}_{Al} + 10^0G^{hcp}_{Sc} + 15^0G^{hcp}_{Zr}$	This work

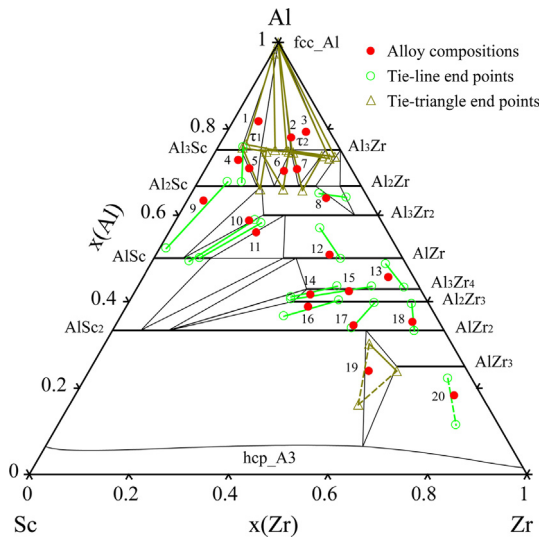

Fig. 7. Calculated isothermal section of the Al-Sc-Zr system at 873 K superimposed with tie lines and tie triangles measured in this work.

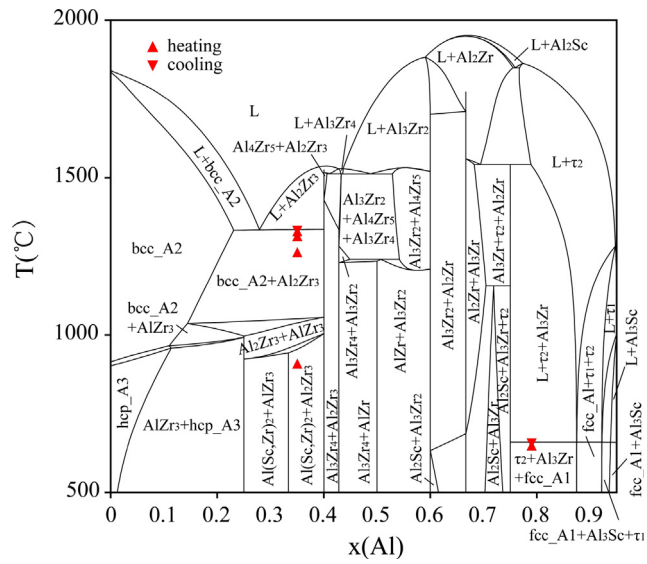

Fig. 8. Calculated vertical section of the Al-Sc-Zr system at 5 at.% Sc.

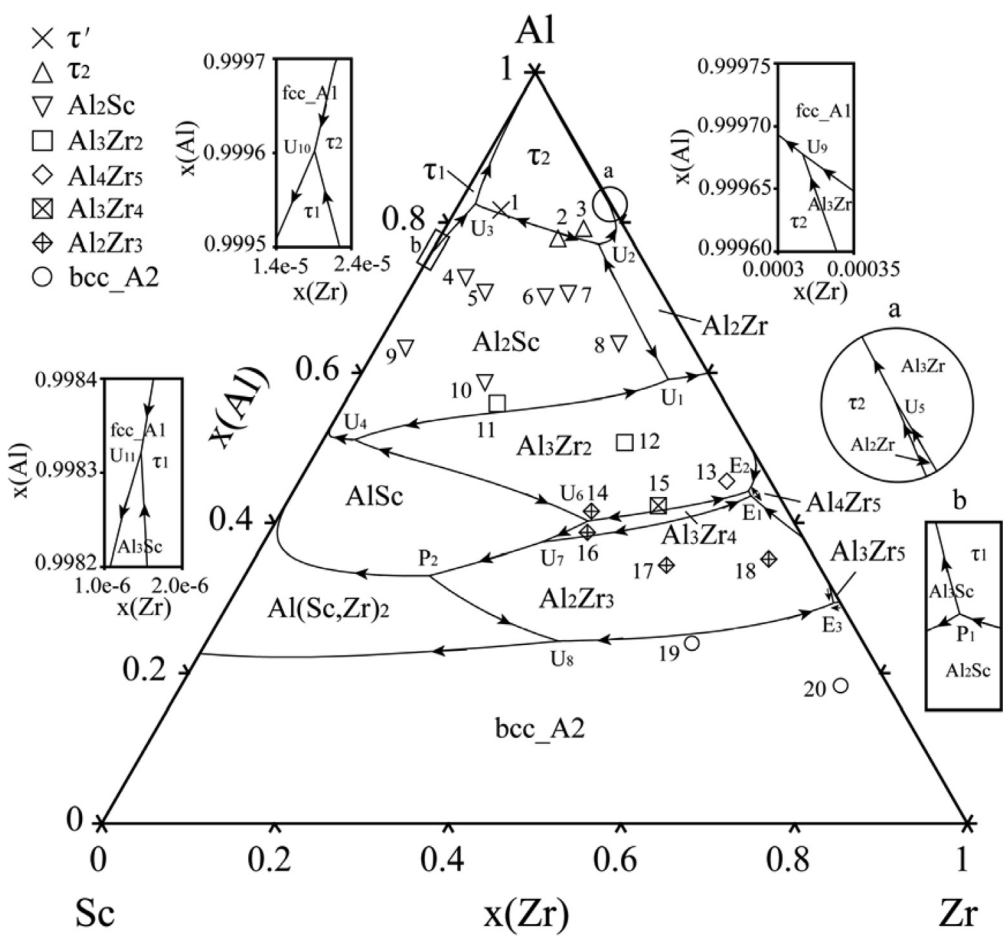

Fig. 9. Calculated liquidus projection of the Al-Sc-Zr system together with the primary crystallization phases observed in the as-cast samples.

Table 5
Calculated invariant reactions of the Al-Sc-Zr system.

Type	Reaction	T (°C)
U_1	Liquid + $Al_2Sc \leftrightarrow Al_2Zr + Al_3Zr_2$	1883
U_2	Liquid + $Al_2Sc \leftrightarrow \tau_2 + Al_2Zr$	1848
U_3	Liquid + $Al_2Sc \leftrightarrow \tau_2 + \tau_1$	1716
U_4	Liquid + $Al_3Zr_2 \leftrightarrow Al_2Sc$ + AlSc	1554
U_5	Liquid + $Al_2Zr \leftrightarrow \tau_2 + Al_3Zr$	1541
U_6	Liquid + $Al_3Zr_2 \leftrightarrow$ AlSc + Al_3Zr_4	1520
E_1	Liquid $\leftrightarrow Al_3Zr_4 + Al_2Zr_3 + Al_4Zr_5$	1511
E_2	Liquid $\leftrightarrow Al_3Zr_4 + Al_4Zr_5 + Al_3Zr_2$	1511
U_7	Liquid + $Al_3Zr_4 \leftrightarrow$ AlSc + Al_2Zr_3	1471
P_1	Liquid + $\tau_1 + Al_2Sc \leftrightarrow Al_3Sc$	1358
E_3	Liquid $\leftrightarrow Al_2Zr_3$ + bcc_A2 + Al_3Zr_5	1339
P_2	Liquid + AlSc + $Al_2Zr_3 \leftrightarrow Al(Sc,Zr)_2$	1335
U_8	Liquid + $Al_2Zr_3 \leftrightarrow Al(Sc,Zr)_2$ + bcc_A2	1236
U_9	Liquid + $Al_3Zr \leftrightarrow \tau_2$ + fcc_A1	661
U_{10}	Liquid + $\tau_2 \leftrightarrow \tau_1$ + fcc_A1	660
U_{11}	Liquid + $\tau_1 \leftrightarrow$ fcc_A1 + Al_3Sc	660

2# and 3#, Al_2Sc for 4–10#, Al_3Zr_2 for 11# and 12#, Al_4Zr_5 for 13#, Al_3Zr_4 for 15#, Al_2Zr_3 for 14# and 16–18#, and bcc_A2 for 19# and 20#. The calculated primary crystallization phases have good agreement with the observed ones expect the samples 2#, 11# and 13–16#, for which the difference is within the experimental error and thus acceptable, as well as the sample 1#, in which the τ' phase is a metastable phase that is not included in the thermodynamic database. Sixteen invariant reactions in total were predicted in this work, as are summarized in Table 5.

6. Conclusions

The isothermal section of the Al-Sc-Zr system at 873 K has been determined using equilibrated alloys. Two stable ternary compounds $Al_{75}Sc_{16}Zr_9$ and $Al_{75}Sc_{10}Zr_{15}$ were newly found to exist in the Al-rich region, for which the crystal structures are not known. According to the experimental results of this work, the Al_6ScZr phase with an $L1_2$ structure could form more easily during solidification than $Al_{75}Sc_{16}Zr_9$ when the content of Al is more than 75 at.% and the atomic ratio of Sc/Zr is over 1, yet it was proved to be a metastable phase. Longer annealing time is needed for the samples in the area of Al-Al_3Sc-$Al_{75}Sc_{10}Zr_{15}$ to reach the equilibrium state in the further experimental investigations. With the discovery of the two stable ternary compounds, the measured phase relations of the Al-rich corner are far different from those reported in the previous study, and so are those of the other regions. Based on all experimental information obtained in this work, the thermodynamic description of the Al-Sc-Zr ternary system was established combining the first-principles calculation and the CALPHAD method, with the Sc-Zr binary system optimized for the first time. The calculated isothermal section at 873 K and the vertical section at 5 at.% Sc have good agreement with the experimental data, and the calculated liquidus projection accounts for most of the primary crystallization phases observed in the as-cast samples satisfactorily. Although some of the phase fields were not evidenced by experimental data, the good accordance between all kinds of calculation results and experimental data has demonstrated the reliability of the present thermodynamic description.

Acknowledgements

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